KASBench Full Disclosure Report

Preamble

This Full Disclosure Report (FDR) documents the results obtained from a single execution of the Kubernetes Autoscaling Benchmark (KASBench) conducted from September 18, 2026 at 20:33:12 UTC to September 19, 2026 at 10:16:10 UTC. In this evaluation, the following autoscaler configurations were tested:

- Kubernetes Horizontal Pod Autoscaler (HPA)
- Kubernetes Vertical Pod Autoscaler (VPA)
- Kubernetes Event-Driven Autoscaler (KEDA)
- No autoscaler (control condition)

The purpose of this report is to completely document the benchmark environment, execution procedures, system configuration, and observed results of a full KASBench execution to enable independent verification and reproduction.

KASBench is a comparative benchmark intended to evaluate Kubernetes autoscaling systems under controlled workloads and documented conditions. It measures autoscaler performance using five metrics (θ^S, θ^A, τ^S, τ^A, and $\bar{C}$) and three composite scores (D^S, D^A, and ζ). These metrics and composite scores measure deviation from the ideal, and are in the range 0 (best) to 100 (worst) inclusive. Lower scores represent autoscaler performance closer to the ideal.

Synchronous underprovisioning penalty (θ^S) quantifies the mean deviation above a pre-determined service level objective (SLO) for synchronous requests initiated by a synthetic load generator over the benchmark period. Results are calculated as a percentage in the range 0 to 100, inclusive, using an upper bound of five-times the SLO. Sub-metrics for response-time (θ^S_{RT}) and failure-rate (θ^S_{FR}) are blended to calculateθ^S.

Synchronous underprovisioned time slices (τ^S) measures the percentage of 30-second time slices during which requests exceed the SLO by more than 20%. It is calculated by blending the sub-metrics for response-time (τ^S_{RT}) and failure-rate (τ^S_{FR}).

Asynchronous underprovisioning penalty (θ^A) quantifies the how long asynchronous tasks wait in queue beyond a predetermined baseline (1 second). Asynchronous tasks are triggered indirectly from synchronous requests. Results are calculated as a percentage in the range 0 to 100, inclusive, using an upper bound of five-times the SLO.

Asynchronous underprovisioned time slices (τ^A) measures the percentage of 30-second time slices during which asynchronous tasks exceed the baseline by more than 20%.

Blended mean resource utilization ($\bar{C}$) is the blended percentage of available CPU and memory requested by each autoscaler in excess of a baseline. The baseline is calculated as the minimum resources that could be allocated by any autoscaler in the test.

The metrics θ^S, τ^S, θ^A, τ^A, and $\bar{C}$ represent deviation from the ideal on a scale of 0 to 100, inclusive.

Derived synchronous composite score (D^S) is the weighted Minkowski deviation (distance) of θ^S and τ^S from zero.

Derived asynchronous composite score (D^A) is the weighted Minkowski deviation (distance) of θ^A and τ^A from zero.

Composite score (ζ) is the weighted Minkowski distance of D^S, D^A, and $\bar{C}$.

The evaluation utilizes a completely random design. Each benchmark evaluation comprises multiple trials. The number of trials per autoscaler is indicated in the *Metadata* section of this document. Trials are independent. Each trial is conducted in an identical, individually built, one-time-use virtual private cloud in a randomly selected Amazon Web Services (AWS) availability zone. Autoscalers under test are randomly assigned to a trial and AWS availability zone.

Autoscalers Under Test

This section describes the configuration of each autoscaler under test. Autoscalers are expected to be configured optimally for the benchmark evaluation, including resource requests and limits, number of replicas, and autoscaler-specific configuration.

Kubernetes Horizontal Pod Autoscaler (HPA)

HPA is a threshold-based horizontal autoscaler. Each deployment is configured with a single replica. Initial CPU and memory requests are established by a calibration process using a fixed workload set at the 15th percentile of benchmark workload. CPU and memory limits were set at 6 cores and 6 GiB, an arbitrarily large number to permit bursting. HPA was configured to allow scaling from 1 to 20 pods. CPU and memory thresholds are set at 70%. The scale down stabilization was set at 75 seconds.

Kubernetes Event-Driven Autoscaler (KEDA)

KEDA is a threshold-based horizontal pod autoscaler driven by events or other metrics. In this evaluation, a hybrid configuration was employed, using KEDA for asynchronous microservices and HPA for others. KEDA monitored the depth of the Apache Kafka queues used for asynchronous processing. The lag threshold was set to 10 items in the queue. KEDA was configured with a minimum replica count of 1 and a maximum of 20 replicas. The polling interval was 15 seconds. The cool down period was set at 75 seconds. For asynchronous services, HPA was configured as described for the HPA-only evaluation.

Kubernetes Vertical Pod Autoscaler (VPA)

VPA is a threshold-based vertical pod autoscaler. VPA initial resource requests and limits were the same as HPA. The minimum CPU was set the same as the requested CPU, as lower amounts were found to lead to instability. The minimum memory was set based on experimentation to avoid fatal out-of-memory errors. Each deployment was configured with two replicas to facilitate vertical scaling with no downtime. The Pod Disruption Budget was set at a minimum of 1 replica.

No Autoscaler (control condition)

As the control condition, evaluations are conducted with no autoscaling configured. Deployments are configured with the same requests and limits as the other tests and a fixed replica count of 1.

The Full Disclosure Report (FDR)

The following sections are produced systematically from data captured during the execution of the benchmark evaluation. All material parameters, settings, versions, hashes, and descriptions are included to foster reproducibility.

Benchmark Metadata

Benchmark Name	KASBench
Version	v1.0
Testing Dates	2026-09-18 20:33:12 UTC to 2026-09-19 10:16:10 UTC
start_time	2026-09-18 20:33:12
end_time	2026-09-19 10:16:10
Reporting Date	2026-10-14 20:26:39 UTC
Region	us-east-1
Benchmark Runner	c6a.large (amd64)
Control Plane	m8i.large (amd64)
Worker Nodes AMD	1 x c6a.4xlarge
Worker Nodes ARM	1 x c6g.4xlarge
AMD Worker AMI ID	ami-0244585558b2aebd7
ARM Worker AMI ID	ami-052bbac83b5bb0ab9
AMD Runner AMI ID	ami-03a891c9de365d954
Kubernetes Version	1.36.1
Autoscalers Evaluated	Kubernetes Horizontal Pod Autoscaler (HPA), Kubernetes Event-Driven Autoscaler (KEDA), Kubernetes Vertical Pod Autoscaler (VPA), No autoscaler (control condition)
Trials Per Autoscaler	4

Software artifacts

Benchmark Components

Component	Image_original	ImageID
KASBench Controller	kasbench/kasbench-controller:0.0.6	sha256:0816294976161de3279756584c5ee4061eafb860d80e436ca0a33dc99cedabc4
KASBench Runner	kasbench/kasbench-runner:0.2.22	sha256:a28c2f8b2a9ca961e2186d0b5e078e373065427b406bc13752a98f9b84d63085
KASBench Load Generator	kasbench/kasbench-load-generator:0.0.21	sha256:5e6ad9d312140a7f6e19f5792aa68b4a1c12d1e2c6999e98f0bd5f24549bca4d

Helm Charts

Name	Version	URL
aws-ebs-csi-driver	2.66.0	https://kubernetes-sigs.github.io/aws-ebs-csi-driver
aws-efs-csi-driver	4.5.0	https://kubernetes-sigs.github.io/aws-efs-csi-driver/
globeco	0.1.51	https://kasbench.github.io/globeco-helm
keda	2.20.2	https://kedacore.github.io/charts
prometheus	29.31.1	https://prometheus-community.github.io/helm-charts

Stateful Services

Stateful Set	Image
postgres	postgres:18.0 docker.io/library/postgres sha256:41fc5342eefba6cc2ccda736aaf034bbbb7c3df0fdb81516eba1ba33f360162c
apache/kafka	apache/kafka:4.0.0 docker.io/apache/kafka sha256:3f7b939115cd4872e9cee9369d80bd69712fde55f9902f46d793f64848dedc75
mongo	mongo:6.0 docker.io/library/mongo sha256:8b6d8f5bbedb25cb73517b65cf99f13aeb75ad5b157a56c479287a840bbad3ac
redis	redis:7-alpine docker.io/library/redis sha256:520775a41a63e77e06c73e35d2fd9cc15921a609516818796b4ecbb813078bc7

Stateless Services

Deployment	Image
globeco-allocation-service	kasbench/globeco-allocation-service-server:1.0.3 docker.io/kasbench/globeco-allocation-service-server sha256:0e6750db5d9e6227554a346f12d0556ea4c62a6047786f2395a539c74218cd5e
globeco-confirmation-service	kasbench/globeco-confirmation-service:1.0.6 docker.io/kasbench/globeco-confirmation-service sha256:5d3267895a231b9210451f253b0ed783301520f036c4eab085b736405e3d7d64
globeco-debug-tools	kasbench/globeco-debug-tools:latest docker.io/kasbench/globeco-debug-tools sha256:9f279ca1ccaf4d118c3de3587bb7946198ac6a14569e3990ea978fdc4f644032
globeco-execution-service	kasbench/globeco-execution-service:1.0.7 docker.io/kasbench/globeco-execution-service sha256:07eb1ec41a44e1b6a80a50b17c4121372bd95a6c51879b08afbe79a28533c9e6
globeco-fix-engine	kasbench/globeco-fix-engine:1.0.13 docker.io/kasbench/globeco-fix-engine sha256:1705dd116250a607d73e1fab8bb6f44f67ea289fa848d4a841d64fe924c1e5d7
globeco-order-generation-service	kasbench/globeco-order-generation-service:1.0.6 docker.io/kasbench/globeco-order-generation-service sha256:a7541d05c1e720573394cb41b443067a05bbb7db75e69e240df1808c5fed5f14
globeco-order-service	kasbench/globeco-order-service:1.0.4 docker.io/kasbench/globeco-order-service sha256:1600b3c42ace32ddf0296468173bc835344e4c885e0b528f4d4e71e51f457b7f
globeco-portfolio-accounting-service	kasbench/globeco-portfolio-accounting-service-server:1.0.2 docker.io/kasbench/globeco-portfolio-accounting-service-server sha256:2b5646e0ccd9a5c4bf4bbce2749f6325c668802175cfc551c36c24fa36dee69d
globeco-portfolio-management-portal	kasbench/globeco-portfolio-management-portal:1.0.5 docker.io/kasbench/globeco-portfolio-management-portal sha256:a65a6022cb244c4baf0eac2ad1f8dddf626ec4ecf0481947e639a0924feb1193
globeco-portfolio-service	kasbench/globeco-portfolio-service:1.0.4 docker.io/kasbench/globeco-portfolio-service sha256:b0561599453606420eb4e904f31214c9c62c5a580c5ca95b25e59f67eca16dd7
	kasbench/globeco-portfolio-service-high-cpu:1.0.4 docker.io/kasbench/globeco-portfolio-service-high-cpu sha256:1a7821e35baf1814f2a7fddbd446ca59b2deaa86fe0a822a5400a27fb59be49c
globeco-pricing-service	kasbench/globeco-pricing-service:1.0.2 docker.io/kasbench/globeco-pricing-service sha256:2509c2a2e69712fa9a559b21a6f820ee537300592e6b77794c7135f282c06dc9
globeco-security-service	kasbench/globeco-security-service:1.0.1 docker.io/kasbench/globeco-security-service sha256:7066be0ea6f76bd60d1ba3c24a30e748c9c767a5eb13a2859204ab765e632241
globeco-trade-service	kasbench/globeco-trade-service:1.0.4 docker.io/kasbench/globeco-trade-service sha256:9c8f610f5fc34dac3eecf396a0f38f9ce4e24c22dea2b1a7acb7726962b26cb4

Raw Performance Metrics

Summarized Performance Metrics by Trial

Trial ID	Autoscaler	Requests	Mean Response Time (ms)	Failures	Failure Rate	Round-trip Completion (%)
trial0001	HPA	271092	100	21	0.008%	1.4%
trial0002	HPA	269018	96	10	0.004%	1.1%
trial0003	VPA	270496	79	11	0.004%	2.2%
trial0004	KEDA	263695	73	168	0.064%	6.2%
trial0005	NONE	259293	205	55	0.021%	1.7%
trial0006	VPA	268839	82	1929	0.718%	2.5%
trial0007	NONE	263339	165	195	0.074%	1.0%
trial0008	HPA	269791	96	21	0.008%	1.1%
trial0009	KEDA	271437	79	4	0.001%	5.9%
trial0010	VPA	269663	75	5	0.002%	2.5%
trial0011	VPA	264706	77	734	0.277%	2.3%
trial0012	NONE	192960	188	16	0.008%	1.8%
trial0013	KEDA	263246	102	130	0.049%	6.8%
trial0014	HPA	270773	84	8	0.003%	1.2%
trial0015	KEDA	270186	94	112	0.041%	6.2%
trial0016	NONE	214979	240	165	0.077%	2.5%

Summarized Performance Metrics by Autoscaler

Autoscaler	Mean Response Time (ms)	Failure Rate	Round-trip Completion (%)
HPA	94	0.006%	1.2%
KEDA	87	0.039%	6.3%
NONE	200	0.045%	1.7%
VPA	78	0.250%	2.4%

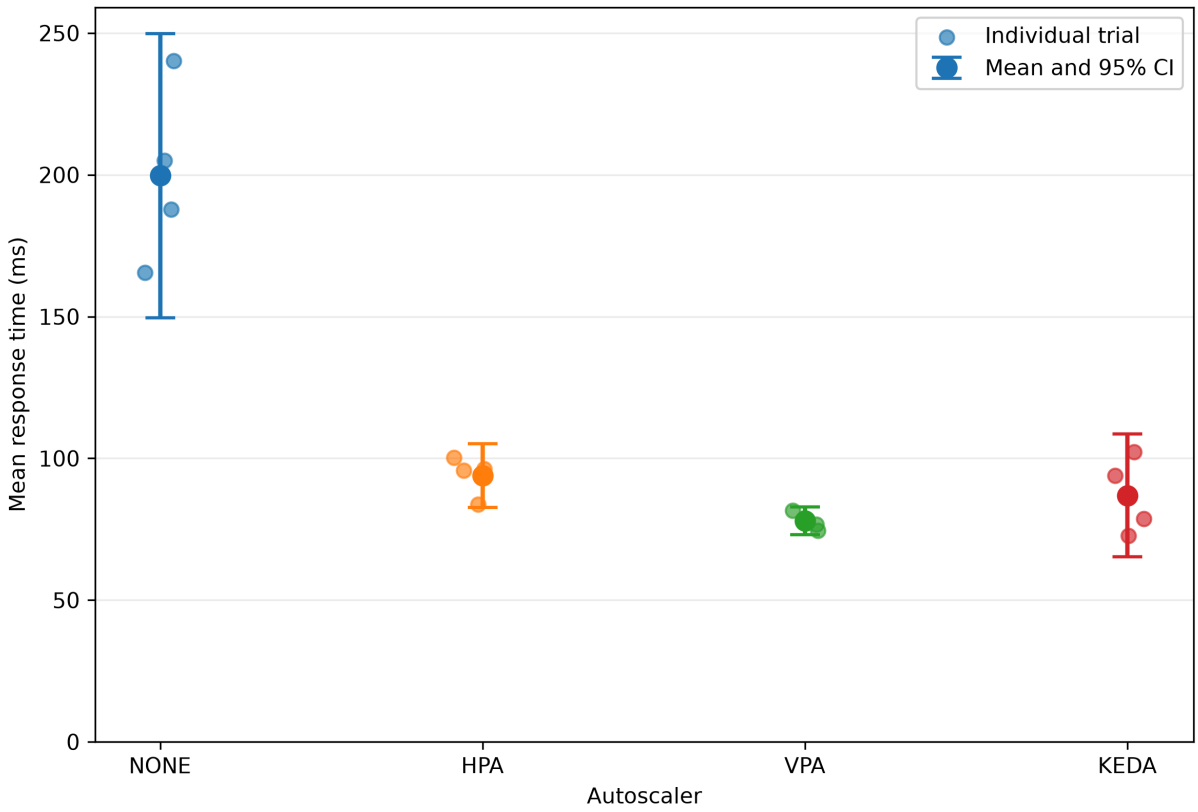


Figure 1. Mean response time by autoscaler.

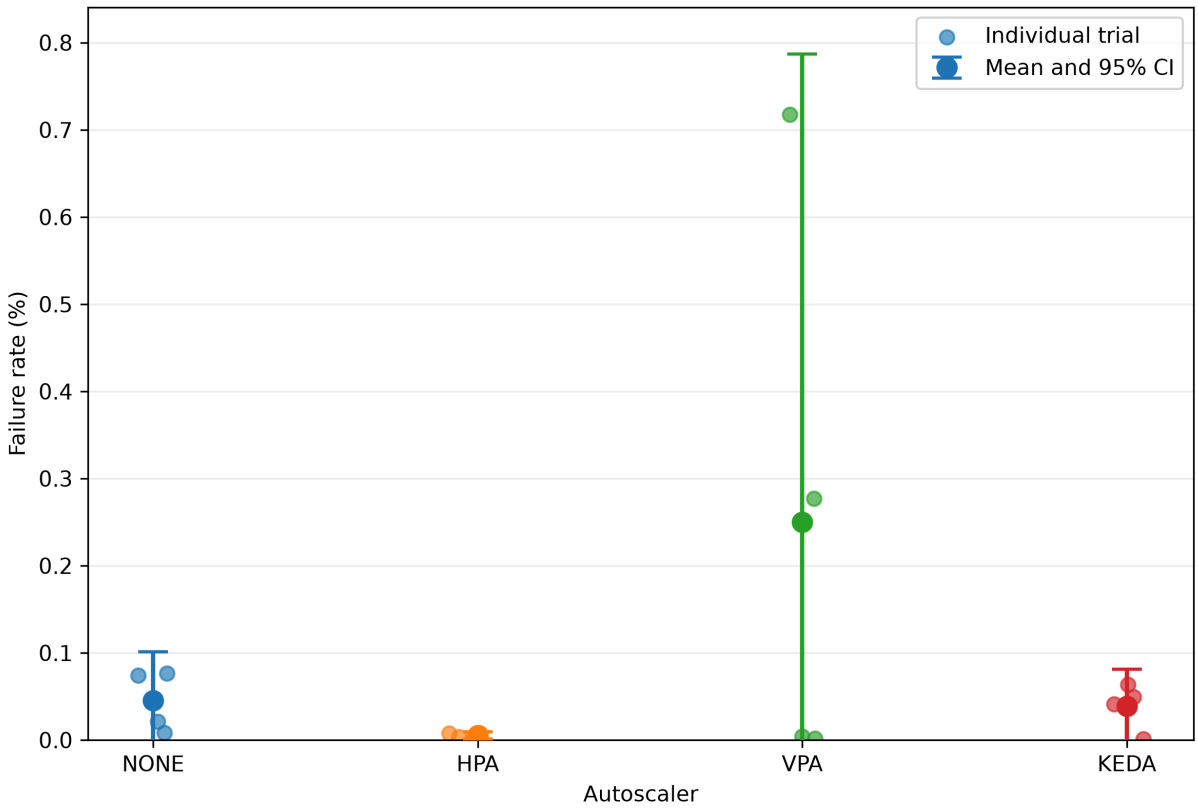


Figure 2. Mean failure rate by autoscaler.

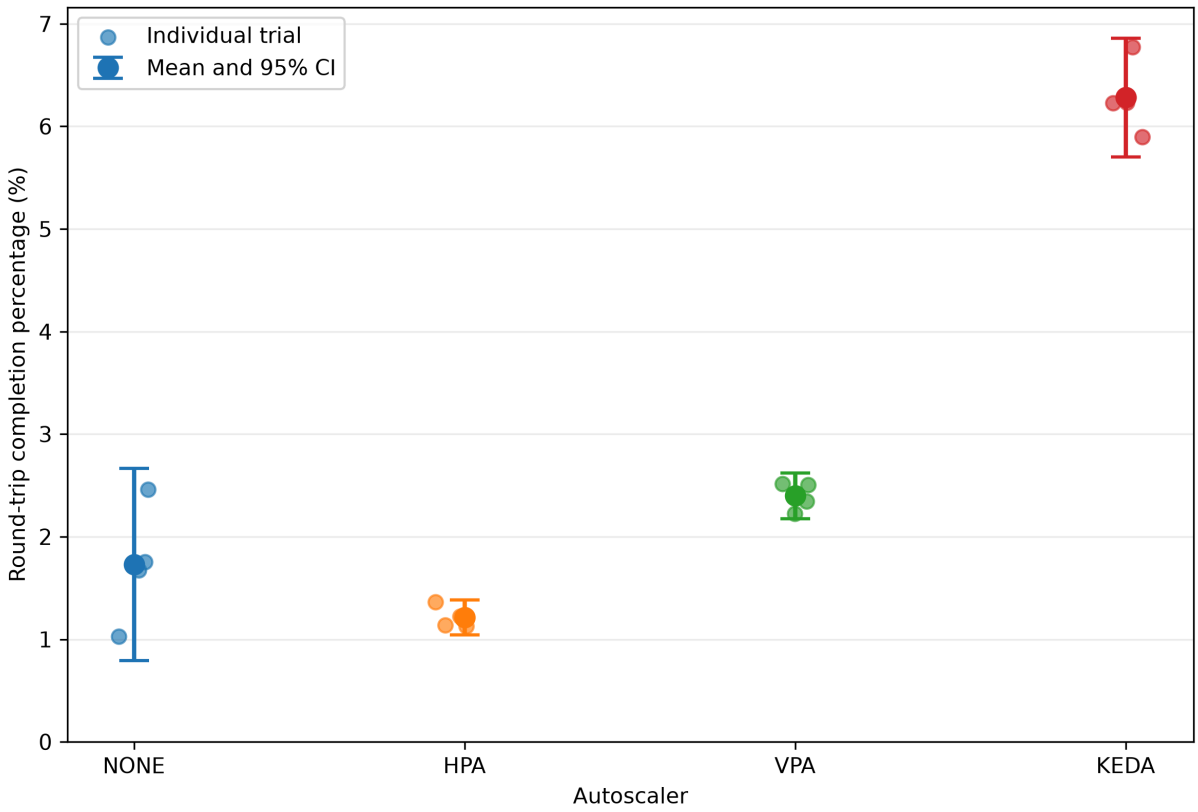


Figure 3. Mean round trip percentage by autoscaler.

Comparative Autoscaler Metrics

Synchronous Underprovisioning Penalty (θ^S) by Autoscaler ($\alpha = 0.5$)

Autoscaler	θ^{RT}	θ^{FR}	θ^S
HPA	40.842	3.549	22.195
KEDA	40.754	7.432	24.093
NONE	70.022	11.060	40.541
VPA	43.294	3.561	23.427

Synchronous Underprovisioned Timeshare (τ^S) by Autoscaler (α = 0.5)

Autoscaler	τ^{RT}	τ^{FR}	τ^S
HPA	88.934	4.918	46.926
KEDA	86.885	8.197	47.541
NONE	92.213	18.443	55.328
VPA	88.115	4.098	46.107

Asynchronous Metrics (θ^A and τ^A) by Autoscaler

Autoscaler	Avg. Lag	θ^A	τ^A
HPA	6817.734	49.580	49.624
KEDA	47.773	13.253	16.787
NONE	3887.296	46.171	46.913
VPA	182.919	32.621	37.184

Cost Metrics by Autoscaler (β = 0.5)

Autoscaler	$\overline{C}_{CPU}$	$\overline{C}_{MEM}$	$\overline{C}$
HPA	48.797	42.140	45.468
KEDA	55.182	46.736	50.959
NONE	13.659	14.592	14.125
VPA	40.956	36.131	38.543

Summary Results by Autoscaler

Autoscaler	θ^S	τ^S	θ^A	τ^A	D^S	D^A	$\overline{C}$	ζ
HPA	22.195	46.926	49.580	49.624	38.515	49.602	45.468	44.892
KEDA	24.093	47.541	13.253	16.787	39.304	15.225	50.959	37.365
NONE	40.541	55.328	46.171	46.913	49.049	46.545	14.125	44.496
VPA	23.427	46.107	32.621	37.184	38.130	35.051	38.543	37.048

Parameters

Parameter	Value
$w_{\theta A}$	0.5
$w_{\tau A}$	0.5
$w_{\theta S}$	0.5
$w_{\tau S}$	0.5
w_S	0.4
w_A	0.4
w_C	0.2
p	3

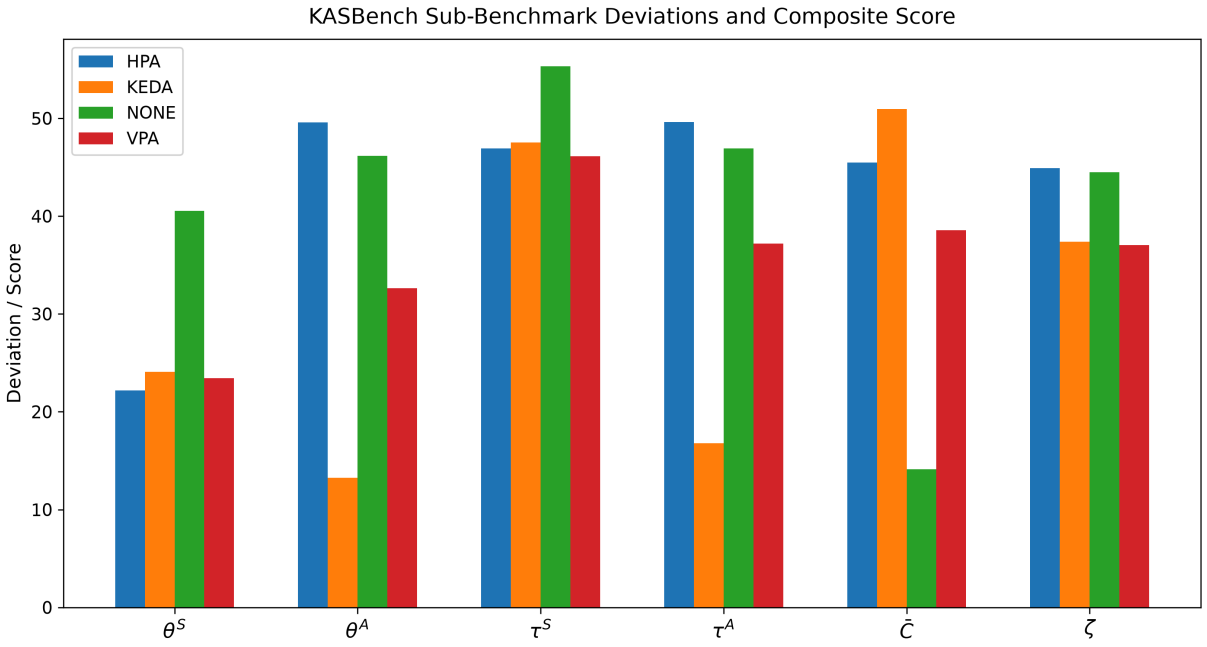


Figure 4. Sub-benchmark deviations and composite score by autoscaler.

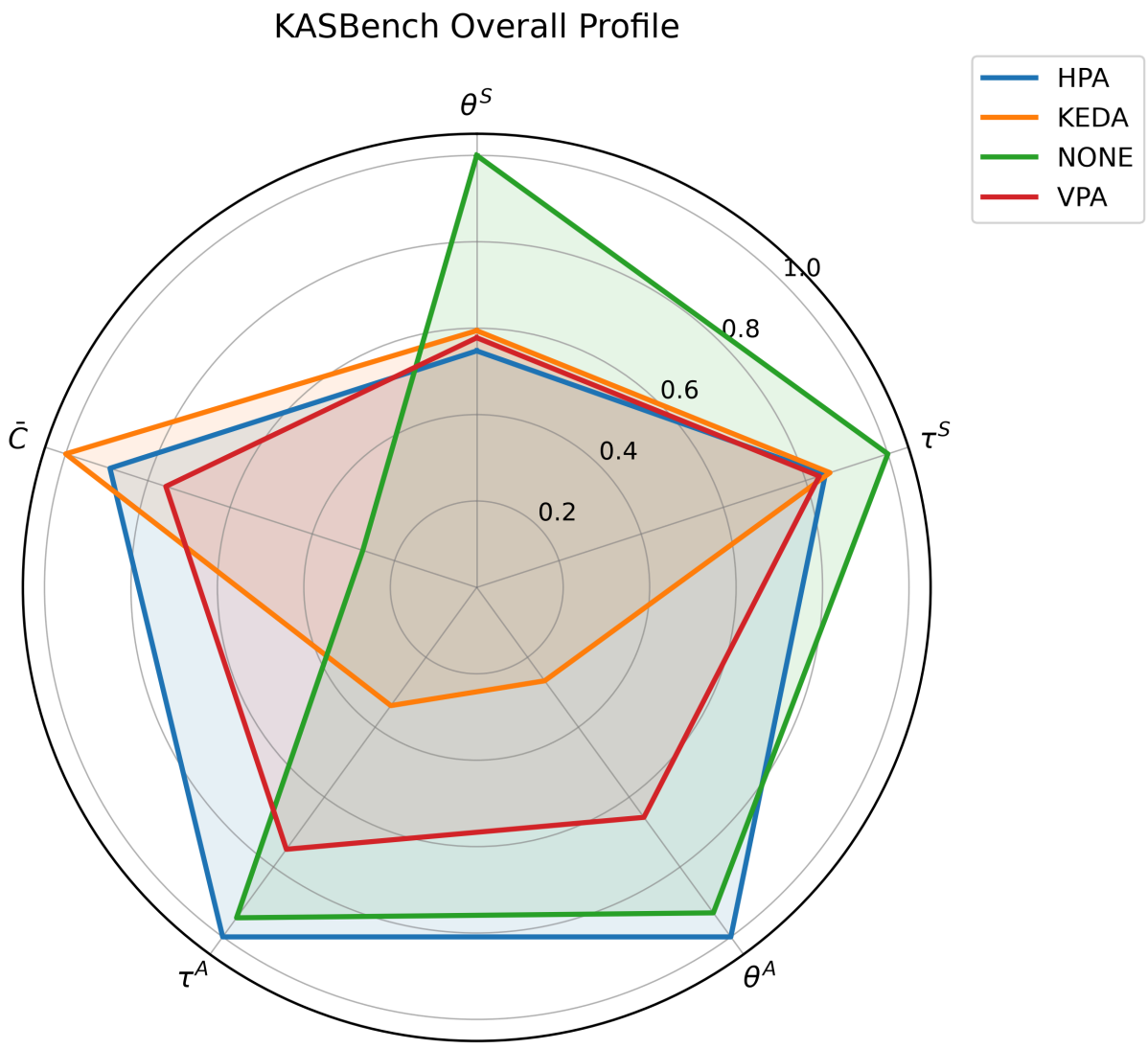


Figure 5. Sub-benchmark deviations by autoscaler.